Proof check request: Theorem 7.4

"Stress identifies the leverage distribution"

Pratik Niroula, Prashant Niroula and Sujan Khatiwada

8 November 2026

What we ask. Please read the proof in Section 2, which is about two pages, and fill in the table in Section 4. The theorem is new. It appears as Theorem 7.4 in the core version of our working paper, "A Sufficient Statistic for Algorithmic Feedback". An AI system (Claude) drafted the proof, and so far only that system has checked it. That is why we ask for a human check. A numerical check (Section 3) passes, but it is not a proof.

1 Setting and statement

Market. Prices follow $x_t = ax_{t-1} + \kappa(F_t + v_t)$ for $t \geq 0$, with $x_{-1} = 0$, $a \in [0, 1)$ and $\kappa > 0$. Here v_t is a probe sequence chosen by the experimenter, and the response (x_t) is observed exactly.

Traders. There are finitely many traders. Trader i has a memory rate $r_i \in [0, 1)$, a capacity $c_i > 0$ and a leverage $\ell_i \in \mathbb{R}$. Its memory state is $m_{i,t} = \sum_{j<t} r_i^{t-1-j} x_j$, and its flow is $c_i \psi(\ell_i m_{i,t})$. Aggregate flow is $F_t = \sum_i c_i \psi(\ell_i m_{i,t})$. The shape ψ is common to all traders and is either tanh or the hard limit $\mathrm{clip}(u) = \max(-1, \min(1, u))$.

Object. The signed capacity measure is $\tilde{\mu} = \sum_i c_i \operatorname{sgn}(\ell_i)\, \delta_{(r_i, |\ell_i|)}$ on $[0, 1) \times (0, \infty)$. Traders with $\ell_i = 0$ contribute nothing.

Theorem 1. *Two populations give identical responses to every probe sequence if and only if* $\tilde{\mu}_P = \tilde{\mu}_{P'}$.

(i) For $\psi = \tanh$*, identical responses to all probes in an arbitrarily small neighbourhood of zero already imply* $\tilde{\mu}_P = \tilde{\mu}_{P'}$.

(ii) For $\psi = \mathrm{clip}$*, probes of all sizes are needed. Probes that keep every memory state inside every band (*$|\ell_i m_{i,t}| \leq 1$*) identify only the kernel* $k_m = \sum_i c_i \ell_i r_i^m$*. The constructive recovery reads the atom at* (r, λ) *from a kink at that atom's band edge.*

A correction already made. An earlier draft of part (ii) said that the traders at leverage $|\ell|$ are identified *only* by probes that reach their own band edge. That is false. When two atoms at the same memory differ and the kernel is fixed, the difference already shows once the higher-leverage atom saturates (Section 3, check T5). The statement above is the corrected one.

2 Proof

Step 1: probes steer the price. F_t is a function of $x_0, \dots, x_{t-1}$. For any target path x^*, the probe $v_t = (x_t^* - a x_{t-1}^*)/\kappa - F_t^P(x_{<t}^*)$ makes population P produce x^*. If P' gives the same response to every probe, it also produces x^* under this v. Then $F_t^P(x_{<t}^*) = (x_t^* - a x_{t-1}^*)/\kappa - v_t = F_t^{P'}(x_{<t}^*)$, so the two flow maps agree on every path. For (i): $|F_t| \le \sum_i c_i |\ell_i| |m_{i,t}|$, so small target paths need only small probes.

Step 2: $\tilde{\mu}$ suffices. ψ is odd, so $c\,\psi(\ell m) = c\,\mathrm{sgn}(\ell)\psi(|\ell| m)$ and $F_t = \int \psi(\lambda\, m_t(r))\, \tilde{\mu}(dr, d\lambda)$, where $m_t(r) = \sum_{j<t} r^{t-1-j} x_j$. So equal $\tilde{\mu}$ give equal flow maps, hence identical closed-loop responses.

Step 3: identification. Let $D = \tilde{\mu}_P - \tilde{\mu}_{P'} = \sum_{k=1}^K d_k \delta_{(r_k, \lambda_k)}$, with distinct atoms and $\lambda_k > 0$. Take $x_0 = \theta u$ and $x_1 = u$. Then $m_2(r) = (r\theta + 1)u$, and equal flow maps at date 2 give

$$g(u) = \sum_k d_k\, \psi(s_k u) = 0, \qquad s_k = \lambda_k(r_k\theta + 1),$$

for all u (clip), or for all u near 0 (tanh). For distinct atoms $k \neq j$, the equation $\lambda_k(r_k\theta + 1) = \lambda_j(r_j\theta + 1)$ is affine in θ and not an identity: if both coefficients matched, then $\lambda_k = \lambda_j$ and $r_k = r_j$. So it has at most one root. Choose $\theta > 0$ outside these finitely many roots. Then the s_k are positive and distinct.

tanh, route A (poles). g is meromorphic on $\mathbb{C}$ and vanishes on a real interval, so it vanishes identically and has zero residue everywhere. Let S be the largest s_k with $d_k \neq 0$. At $u^* = i\pi/(2S)$, $\tanh(Su)$ has a simple pole with residue $1/S$. For $s_k < S$, $s_k u^* = i\pi s_k/(2S)$ lies strictly between 0 and $i\pi/2$ on the imaginary axis, which is not a pole. So $\mathrm{Res}_{u^*} g = d_k/S \neq 0$, a contradiction.

tanh, route B (Taylor). $\tanh u = \sum_{n\ge 1} \tau_{2n-1} u^{2n-1}$ with $\tau_{2n-1} = 2^{2n}(2^{2n} - 1)B_{2n}/(2n)! \neq 0$. If $g \equiv 0$ near 0, then $\sum_k (d_k s_k)(s_k^2)^{n-1} = 0$ for $n = 1, \dots, K$, a Vandermonde system in the distinct numbers s_k^2. So $d_k = 0$.

clip. On $u \ge 0$, the function $\mathrm{clip}(s_k u)$ has slope s_k on $[0, 1/s_k)$ and slope 0 after. Since the s_k are distinct, the slope of g changes at $u = 1/s_k$ by exactly $-d_k s_k$. If $g \equiv 0$, every slope change vanishes, so $d_k = 0$. The change at $1/s_k$ is visible only on paths with $u \ge 1/s_k$.

Necessity part of (ii). This is Theorem 7.2(iv) of the paper. While every state is inside every band, both populations trade linearly with the same kernel, so their responses coincide.

3 Numerical check (not a proof)

The script `research_extension/nonlinear_scoping/thm74_numeric_check.py` (also packaged as the Colab notebook `Thm74_Check_Colab.ipynb`) runs five checks for three seeds:

- **T1.** Same $\tilde{\mu}$, after random splits and offsetting pairs, gives identical responses. Largest gap: 1.4×10^{-14} over 40 populations and 200 random probes, under both shapes.
- **T2.** Same kernel, different $\tilde{\mu}$: under clip, identical below pulse size 5, then separated. Under tanh, the first-tap gap scales with fitted exponent 2.98, against the predicted 3.
- **T3.** Constructive recovery of $\tilde{\mu}$ under clip from two directions recovers all atoms in 25 of 25 random populations. Atoms were first matched by exact rounded keys, and 3 cases failed by one unit in the third decimal (the grid step). With a tolerance equal to the grid step, all 25 match.

- **T4.** Cubic Taylor coefficient of the tanh flow-map difference: fitted 9.2264061×10^{-4}, predicted 9.2264063×10^{-4}.
- **T5.** The counterexample to the old wording of (ii): the gap is 0 inside all bands and 0.375 between the two band edges.

4 Questions and response table

1. Is Step 1 (steering) valid as stated, including the small-probe version used for (i)?
2. Is the choice of θ in Step 3 correct, and does the reduction to $g(u) = 0$ lose anything?
3. Is route A correct? In particular, is it correct that no smaller s_k has a pole at u^*? Is route B correct? Is either route preferable?
4. Is the clip argument complete? Is the corrected wording of (ii) accurate?
5. Is $\tilde{\mu}$ exactly the equivalence class, that is, is Step 2 complete?
6. Is there an odd analytic ψ for which the result fails? We expect failure when ψ has only finitely many nonzero odd Taylor coefficients, for example a cubic, because route B then gives only finitely many moments.

Question	Correct / gap / error	Comment
1		
2		
3		
4		
5		
6		

Reader: _____ Date: _____ We will acknowledge the reader by name in the paper only with their permission.